

TRUST-ENHANCED GRAPH NEURAL NETWORKS FOR
TRANSPARENT RECOMMENDATIONS

Anonymous authors
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ABSTRACT

In the evolving landscape of digital platforms, the demand for robust recommendation systems is paramount to manage the deluge of user-generated data. Graph Neural Networks (GNNs) have emerged as a potent strategy in recognizing intricate user-item interactions due to their ability to leverage structural data insights. However, existing GNN-based models often overlook trust dynamics, a critical factor in ensuring recommendation reliability and transparency. Despite recognition of trust's potential to address biases and enhance models' interpretability, its integration with sophisticated network-based techniques remains under-explored. Responding to this gap, we propose the Trust-Enhanced Graph-Based Recommendation Model (GTERM), which seamlessly incorporates trust metrics within the GNN framework. GTERM transforms raw interaction data into a trust-augmented graph, employing graph convolutional and attention mechanisms to emphasize trust-enriched interactions, thereby refining recommendation accuracy and transparency. The proposed model achieves notable improvements over baseline methods, as evidenced in diverse experimental evaluations, demonstrating its capacity to deliver more accurate, trustworthy, and interpretable recommendations. Through the integration of trust factors, GTERM fosters user acceptance and enhances system performance by resolving key challenges related to the lack of interpretability and trustworthiness in traditional GNN-based systems.

1 PROPOSED METHOD: TRUST-ENHANCED GRAPH-BASED
RECOMMENDATION MODEL (GTERM)

The rapid expansion of digital platforms has resulted in an abundant accumulation of user-generated data, underscoring the demand for efficient recommendation systems capable of filtering this information to deliver personalized content. Graph Neural Networks (GNNs) have gained prominence as an effective tool for these systems, particularly due to their ability to model complex user-item interactions through graph structures (Wu et al., 2019; Kipf & Welling, 2017). GNNs excel in capturing both connectivity and complex interaction patterns, thus providing a robust framework for enhancing recommendations through structural insights. Despite these advancements, GNNs often neglect the critical role of trust dynamics in recommendation scenarios (Veličković et al., 2018).

Research on trust mechanisms in recommendation systems has highlighted their potential to enhance recommendation reliability by incorporating user trust scores (Massa & Avesani, 2007; Guo et al., 2017). These trust-based approaches aim to mitigate biases from unreliable interactions and increase transparency by elucidating the rationale behind recommendations. Nevertheless, many trust-based methodologies operate independently of sophisticated network-based techniques, missing the opportunity to integrate the comprehensive interaction modeling capabilities of GNNs (Konstas et al., 2009; Ma et al., 2011). The incorporation of trust into GNN frameworks remains inadequately explored, presenting an opportunity for innovation that our proposed model seeks to exploit.

The integration of trust and GNNs faces several challenges. Conventional GNN-based recommendation models frequently lack explicit trust metrics, limiting their capacity to deliver interpretable and user-focused recommendations. Specifically, while GNNs adeptly model intricate interaction structures, they may not distinguish interactions based on their trustworthiness. Moreover, the opacity of such models often compromises user confidence and acceptance (Zhang et al., 2014). Addressing

 **Meta Review Status:** The meta review and decision have not been released yet.

AI Review from DeepReviewer

 AI Review from DeepReviewer will be automatically processed

Summary

This paper introduces the Trust-Enhanced Graph-Based Recommendation Model (GTERM), a novel approach that integrates trust metrics into a Graph Neural Network (GNN) framework to enhance the accuracy and transparency of recommendation systems. The core idea behind GTERM is to construct a trust-augmented graph from user-item interactions, where edges are weighted by trust scores derived from historical interaction metrics. This graph is then processed using a hybrid GNN architecture that combines Graph Convolutional Networks (GCN) and Graph Attention Networks (GAT). The GCN layers aggregate neighborhood information, while the GAT layers prioritize trustworthy interactions through attention mechanisms. The model outputs both a recommendation score and a path-based explanation, aiming to provide users with more transparent and interpretable recommendations. The authors evaluate GTERM on several datasets, comparing its performance against Collaborative Filtering (CF) and standard GCN models. The results suggest that GTERM outperforms these baselines in terms of recommendation accuracy and interpretability. The paper's main contribution lies in its attempt to address the limitations of traditional GNN-based recommendation systems by explicitly incorporating trust dynamics, thereby aiming to improve both the reliability and transparency of the recommendations. However, as I will detail, the paper suffers from several methodological and experimental shortcomings that limit the impact of its contributions. The lack of detail in the methodology, the limited scope of the experimental evaluation, and the poor presentation of the results all contribute to a less convincing argument for the proposed model. Despite the promising concept, the paper requires significant improvements to fully realize its potential.

Strengths

The most compelling strength of this paper lies in its attempt to address a significant gap in the field of recommendation systems by integrating trust mechanisms with Graph Neural Networks (GNNs). The idea of incorporating trust metrics into a GNN framework is innovative and tackles a critical need for more reliable and transparent recommendations. By constructing a trust-augmented graph, the authors aim to emphasize trust-enriched interactions, which is a valuable contribution to the field. The use of both GCN and GAT layers is also a positive aspect, as it allows the model to aggregate neighborhood information while also prioritizing trustworthy interactions through attention mechanisms. Furthermore, the output of the Trust-Integrated Recommendation Framework, which provides path-based explanations for each recommendation, is commendable. This feature enhances the interpretability of the recommendations, which is a crucial aspect for user trust and acceptance. The model's ability to provide clear and understandable explanations for the suggested items is a significant step towards more transparent recommendation systems. The authors also plan to release the code, which promotes the reproducibility of the research results, which is a positive step for the research community. Despite these strengths, the paper is significantly undermined by several weaknesses that need to be addressed.

Weaknesses

After a thorough examination of the paper, I have identified several significant weaknesses that undermine its overall credibility and impact. Firstly, the paper lacks sufficient detail in its methodology, particularly concerning the hyperparameter tuning process. The trust score balancing parameter, α , is initially set to 0.5, and the authors mention adjusting it during hyperparameter tuning. However, they fail to provide a clear methodology for this optimization. The paper does not specify the range of values explored for α , the optimization strategy employed (e.g., grid search, random search), or the criteria used to select the optimal value. This lack of detail makes it difficult to assess the robustness of the model and its sensitivity to different values of α . The absence of this information makes it impossible to replicate the results and understand the model's behavior. This is a critical oversight that significantly impacts the reproducibility and reliability of the findings. My confidence in this weakness is high, as the paper explicitly mentions the tuning process but provides no details on how it was conducted. Secondly, the paper suffers from a lack of a comprehensive comparative analysis. While the authors compare GTERM against Collaborative Filtering (CF) and Graph Convolutional Networks (GCN), they fail to include comparisons with other relevant state-of-the-art models that incorporate trust and attention mechanisms. Specifically, models such as TrustSVD and Graph Attention Networks (GAT) are not included in the experimental comparison, despite being mentioned in the related work section. This limited comparison makes it challenging to fully understand the advantages and disadvantages of the proposed model relative to existing approaches. The absence of these comparisons makes it difficult to assess the true novelty and effectiveness of GTERM. My confidence in this weakness is high, as the paper explicitly states the baselines used and the results table confirms the limited comparison. Thirdly, the paper does not provide a detailed explanation or derivation of the formulas used in the GCN and GAT layers. The equations presented in sections 3.2 and 3.3 lack proper referencing and explanation of their origins and adaptations. While the paper cites foundational works for GCN and GAT, it does not explicitly detail if these are used directly or if any modifications were made to adapt them for the trust-enhanced framework of GTERM. This lack of clarity hinders the reproducibility and understanding of the proposed methodology. The absence of specific derivations or explanations of adaptations makes it harder to fully understand the implementation and hinders reproducibility. My confidence in this weakness is high, as the paper presents the formulas but lacks detailed explanation of their derivation or adaptation within the GTERM framework. Fourthly, the paper's formatting is problematic, with text in the main body appearing significantly reduced in size compared to the template, suggesting potential issues with the document's layout or compilation. This formatting issue makes the paper difficult to read and does not adhere to standard academic formatting. My confidence in this weakness is high, as it is based on a visual inspection of the provided paper. Fifthly, the overall writing quality is poor, with numerous grammatical errors, awkward phrasing, and unclear sentences, making it difficult to understand the authors' arguments and contributions. For example, in the Abstract, the sentence "In the evolving landscape of digital platforms, the demand for robust recommendation systems is paramount to manage the deluge of user-generated data" could be phrased more clearly. In the Introduction, the sentence "GNNs excel in capturing both connectivity and complex interaction patterns" contains a typo ("paterns"). In the Method section, the sentence "Efficient Processing Adaptation: PyTorch Geometric DataLoader partitions graph data into manageable subsets, optimizing memory and computational efficiency for GNN workflows" has spacing and spelling errors ("efciency"). These errors detract from the paper's credibility and make it difficult to follow the authors' reasoning. My confidence in this weakness is high, as multiple examples of grammatical errors and awkward phrasing are present throughout the paper. Sixthly, the paper lacks a clear and well-defined research motivation. While the "Motivation" section discusses the importance of recommendation systems and the limitations of traditional methods, it could be more explicit in articulating the specific gap that GTERM uniquely addresses and why this particular solution is necessary. The connection between these points and the specific need for *this* proposed solution (GTERM) could be more explicitly stated. My confidence in this weakness is medium, as the motivation section identifies problems but could more clearly connect them to the specific need for GTERM. Seventhly, the methodology section is insufficiently detailed, lacking clear explanations of the proposed approach, algorithms, and technical processes involved in the research. While the "Method" section provides a high-level overview, certain aspects lack depth. For example, the "Trust Metric Derivation" mentions calculating trust scores from historical interaction metrics but doesn't specify the exact formula or method used. The "GNN-Based Interaction Prediction" describes the GCN and GAT layers but doesn't detail the specific configurations or adaptations. My confidence in this weakness is high, as there is a lack of specific details in "Trust Metric Derivation" and "GNN-Based Interaction Prediction." Eighthly, the experimental design is flawed, with inadequate descriptions of the datasets, evaluation metrics, and baseline comparisons. The "Datasets" subsection lists the datasets used but provides minimal details about their characteristics (e.g., number of users, items, interactions). The "Evaluation Metrics" lists Recall@10, Precision@10, and NDCG@10, which are standard but lacks justification for their choice. The "Baseline Comparisons" lists CF and GCN but lacks justification for these specific choices and doesn't discuss why other relevant baselines were excluded. My confidence in this weakness is high, as there is a lack of dataset details, metric justification, and baseline rationale in the "Experiments" section. Ninthly, the results and analysis sections are poorly presented, with unclear interpretations of the findings and a lack of insightful discussion regarding the implications of the research. The "Results and Analysis" section presents tables

comparing GTERM with baselines and an ablation study. The "Analysis" subsection provides some interpretation, stating that GTERM outperforms baselines and that trust integration and attention mechanisms are important. However, the depth of analysis could be greater. For example, the "Error Analysis" is very brief. My confidence in this weakness is medium, as the analysis is present but lacks depth and detailed error analysis. Tenthly, the conclusion is weak and does not effectively summarize the research or highlight the key contributions and potential impact of the work. The "Conclusion" section summarizes the work by stating the problem addressed, the proposed solution (GTERM), and the key findings (improved accuracy and interpretability). It also mentions future work. However, it could be more impactful by explicitly reiterating the novelty of the approach and its potential broader implications. My confidence in this weakness is medium, as the conclusion is present but could be more impactful. Finally, the paper does not explain how the trust metric is calculated. The paper mentions that trust scores are derived from historical interaction metrics like review frequencies and ratings, but it does not provide the specific calculation method or formula. This lack of detail makes the trust metric a "black box." My confidence in this weakness is high, as the paper mentions the data used for trust calculation but not the calculation method. Furthermore, the paper does not explain how to derive explanations from the model's architecture. The paper claims that path-based explanations are generated, but it lacks a detailed explanation of the process or algorithm used to derive these explanations from the GNN architecture. My confidence in this weakness is high, as the paper mentions path-based explanations but lacks details on their derivation. Lastly, the experimental results are not fully described, as the dataset statistics are not provided, and the baselines are too simple. The "Datasets" subsection lists the datasets used but does not provide statistics like the number of users, items, and interactions. The "Baseline Comparisons" subsection lists CF and GCN as baselines, which could be considered relatively simple compared to more recent GNN-based recommendation models. My confidence in this weakness is high, as there is a lack of dataset statistics and the simplicity of the chosen baselines.

Suggestions

To address the identified weaknesses, I recommend several concrete and actionable improvements. Firstly, the authors must provide a comprehensive description of the hyperparameter tuning process for the trust score balancing parameter α . This should involve specifying the range of values explored for α , the optimization strategy employed (e.g., grid search, random search, Bayesian optimization), and the specific criteria used to evaluate the performance at each step. Furthermore, a sensitivity analysis of the model's performance with respect to different values of α should be included. This analysis should include a plot or table showing how the model's performance metrics (e.g., precision, recall, NDCG) change as α varies. This would provide a more robust understanding of the model's behavior and the optimal setting for α . The authors should also clarify whether the optimization was performed on a validation set and how the validation set was constructed to avoid overfitting. Secondly, the authors should include a broader range of state-of-the-art models in their experiments. This should include models that incorporate trust mechanisms, such as TrustSVD, as well as models that utilize attention mechanisms, such as Graph Attention Networks (GAT). The comparison should not only focus on overall performance metrics but also on the interpretability and transparency of the recommendations. The authors should provide a detailed analysis of the strengths and weaknesses of the proposed model relative to each baseline, highlighting the specific scenarios where the proposed model performs better or worse. This would provide a more comprehensive understanding of the model's capabilities and limitations. Furthermore, the authors should consider using more diverse datasets to evaluate the model's generalizability. Thirdly, the authors should provide a detailed explanation and derivation of the formulas used in the GCN and GAT layers. This should include a clear articulation of the theoretical underpinnings of these equations, including any modifications made to existing formulations. The authors should also clearly reference the source of each equation and explain how it was adapted for the specific context of their model. For example, if the GCN layer equation is based on existing work, the authors should explicitly state this and cite the relevant paper. Similarly, for the GAT layer, the authors should explain how the attention mechanism is applied and how the trust metrics are integrated into the attention weights. This would enhance the reproducibility and understanding of the proposed methodology. Fourthly, the authors need to correct the formatting issues, particularly the reduced text size in the main body, to ensure readability and adherence to standard academic formatting. This may involve revisiting the LaTeX compilation process or using a different typesetting tool. Fifthly, the writing quality must be substantially improved. This includes addressing grammatical errors, awkward phrasing, and unclear sentences. The authors should consider using a professional editing service or seeking assistance from a native English speaker to refine the language and ensure clarity. Sixthly, the paper needs a clear and compelling research motivation. The authors should explicitly state the problem they are addressing, why it is important, and how their proposed solution contributes to the field. This should be done in the introduction section, setting the stage for the rest of the paper. Seventhly, the methodology section requires a complete rewrite, providing detailed explanations of the proposed approach, algorithms, and technical processes. This should include clear descriptions of the GNN architecture, the trust metric integration, and any other relevant technical details. The authors should also provide a clear explanation of how

the model is trained and evaluated. Eighthly, the experimental design needs to be thoroughly revised. The authors should provide detailed descriptions of the datasets used, including their size, characteristics, and any preprocessing steps. They should also clearly define the evaluation metrics used and justify their choice. Furthermore, the baseline comparisons should be carefully selected to provide a meaningful assessment of the proposed model's performance. Ninthly, the results and analysis sections need to be improved by providing clear interpretations of the findings and a detailed discussion of their implications. The authors should also discuss the limitations of their approach and suggest directions for future research. Finally, the conclusion should be strengthened to effectively summarize the research, highlight the key contributions, and discuss the potential impact of the work. The authors should also consider adding a section on ethical considerations, given the use of trust metrics in recommendations. Tenthly, the paper needs to provide a clear and detailed explanation of how the trust metric is calculated. Simply stating that trust scores are derived from 'review frequencies and ratings' is insufficient. The specific formula or algorithm used to combine these factors into a single trust score needs to be explicitly stated. For example, are review frequencies and ratings combined linearly, or is a more complex function used? Are there any weighting factors applied to either review frequency or rating? Furthermore, how are edge cases handled, such as users with no reviews or items with no ratings? Without this level of detail, the trust metric remains a black box, making it difficult to assess the validity of the approach and reproduce the results. The authors should also clarify whether the trust metric is static or dynamically updated during the training process. If it is static, this should be explicitly stated, and the rationale for this choice should be provided. If it is dynamic, the update mechanism needs to be described in detail. Eleventhly, regarding the derivation of explanations, the paper needs to provide a concrete mechanism for how path-based explanations are generated from the GNN architecture. The current description is too vague. It is not clear how the model identifies the 'most influential paths' or how these paths are translated into human-understandable explanations. For example, are these paths simply the shortest paths between user and item nodes, or are they weighted based on the trust scores or other factors? The authors should provide a specific algorithm or procedure for extracting these paths. Furthermore, it is important to clarify how the model handles cases where multiple paths exist between a user and an item. Does it select the most influential path, or does it provide a set of potential explanations? The paper should also discuss the computational complexity of generating these explanations, as this could be a limiting factor in practical applications. A clear example of how a path-based explanation is presented to the user would also be beneficial. Finally, the experimental section needs to be significantly improved. The dataset statistics should be provided, including the number of users, items, and interactions, as well as the sparsity of the data. The baselines used for comparison are too simplistic. The authors should compare their model against more recent and competitive GNN-based recommendation models, such as LightGCN or other state-of-the-art methods. The evaluation metrics should also be clearly defined, and the results should be presented with statistical significance. It is also important to clarify whether the reported results are based on a single run or multiple runs, and if multiple runs are used, the standard deviation should be reported. The authors should also consider performing ablation studies to evaluate the contribution of each component of their model, such as the trust-augmented graph and the attention mechanism. This would provide a more comprehensive understanding of the model's performance and the effectiveness of the proposed approach.

? Questions

Based on my analysis, I have several key questions that I believe are crucial for understanding the paper's methodology and results. Firstly, how was the trust score balancing parameter α tuned, and what impact does its variation have on the model's performance? The paper mentions that α was initially set to 0.5 and adjusted during hyperparameter tuning, but it lacks a detailed explanation of the tuning process and the criteria used for optimization. A more thorough description of how α was optimized and its effect on the model's performance would provide valuable insights into the model's sensitivity and robustness. Secondly, can the authors provide a more detailed comparison of their model with a broader range of state-of-the-art models, including those that incorporate trust and attention mechanisms? The current comparison is limited to Collaborative Filtering (CF) and Graph Convolutional Networks (GCN). Expanding the range of models for comparison, such as TrustSVD and other GAT-based models, would better demonstrate the advantages and potential limitations of the proposed model. Thirdly, the paper lacks clarity on the formulas used in the GCN and GAT layers. Could the authors provide more detailed explanations or derivations of these equations, including references to their origins and any adaptations made? Clearer articulation of the theoretical underpinnings of these equations would enhance the paper's reproducibility and understanding. Fourthly, what is the specific formula or algorithm used to calculate the trust metric from review frequencies and ratings? The paper mentions that trust scores are derived from these metrics, but it does not provide the specific calculation method. A clear explanation of this process is essential for understanding the model's behavior. Fifthly, how are path-based explanations generated from the GNN architecture? The paper claims that these explanations are derived from graph structures, but it does not detail the specific mechanism or algorithm used to identify and generate these

explanations. A clear explanation of this process is crucial for evaluating the model's interpretability. Finally, what are the specific characteristics of the datasets used in the experiments, including the number of users, items, and interactions? The paper lists the datasets but does not provide detailed statistics. This information is necessary for assessing the generalizability of the results. These questions target the core methodological choices and experimental design of the paper, and addressing them would significantly improve the paper's clarity and credibility.

Scores

Soundness: **1.0**

Presentation: **1.0**

Contribution: **1.0**

Rating: 2.0

AI Review from ZGCA

 AI Review from ZGCA will be automatically processed

Summary

The paper proposes GTERM, a trust-enhanced graph-based recommendation model that integrates trust metrics into a GNN framework. The approach (i) constructs a trust-augmented user–item bipartite graph where trust scores derived from historical interactions (e.g., review frequency, ratings) are encoded as edge weights (Section 3.1), (ii) learns embeddings via a hybrid GCN + GAT architecture to emphasize trust-influenced interactions (Section 3.2), and (iii) produces final recommendation scores via a composite function that linearly combines a trust score and a GNN interaction score with a tunable alpha parameter (Section 3.2). The authors also claim a transparent recommendation framework that outputs path-based explanations (Section 3.3). Experiments on Yelp, Gowalla, ML-10M, and Amazon Books report improvements over CF and a standard GCN in Recall@10, Precision@10, and NDCG@10, with simple ablations for removing trust and attention (Section 4).

Strengths

- Addresses an important and timely problem: incorporating trust and transparency into GNN-based recommendation (Abstract; Sections 1, 2.2, 2.3).
- Conceptually simple integration pipeline: encode trust as edge weights, use attention to prioritize higher-trust edges, and combine with a composite score (Sections 3.1–3.3).
- Ablation indicates both trust integration and attention contribute positively (Table 2).

- Claims improvements on multiple datasets (Section 4.1), with statistically significant gains over CF and a baseline GCN (Table 1; Section 4.3).

✗ Weaknesses

- The paper lacks a precise, reproducible algorithm for computing the trust scores that are central to the method. Section 3.1 states trust is derived from 'historical interaction metrics (e.g., review frequencies, ratings)' but provides no formulae, normalization, calibration, temporal handling, or validation. This undermines the main technical claim and reproducibility.
- Training objective and learning details are unspecified: no loss function, negative sampling strategy, or whether the composite score is used during training vs only at inference (Section 3.2).
- Transparency/interpretability claims are not evaluated. Section 3.3 promises path-based explanations, yet there is no quantitative (e.g., fidelity, plausibility, coverage) or qualitative evaluation (no example explanations) in Section 4.
- Baselines are limited. Comparisons only to CF and a 'standard GCN' underrepresent current recommenders and trust-aware methods (e.g., TrustSVD) and modern GNN recommenders (e.g., NGCF/LightGCN/GraphRec/SocialGCN).
- Reporting is incomplete: Table 1 appears to give one set of numbers without per-dataset breakdowns; no standard deviations or confidence intervals; t-test reported as 'p; 0.05' (Section 4.3). Data splits, random seeds, training epochs, embedding sizes, optimizer, weight decay, early stopping, and hardware are missing (Sections 4.1–4.2).
- Methodological clarity issues: Section 3.2 mentions 'global pooling' to obtain a graph representation for interaction probabilities, which is atypical for user–item link prediction and needs justification. The 'Trust-weighted interaction prioritization' subsection (3.2.1) is only a single sentence without technical detail.
- Some presentation and citation quality issues (e.g., placeholders '(?)', duplicated 'Tables 4.3 and 4.3', minor typographical errors), detracting from clarity.

? Questions

- Trust derivation: Please provide the exact algorithm for computing trust scores. What is the unit of trust (user–item, user–user, item–item)? Include formulae, normalization, handling of sparsity/cold-start, temporal decay, and any calibration to avoid popularity bias.
- Training objective: What loss function do you optimize (e.g., BPR, cross-entropy)? Are trust-weighted edges used in message passing only, or also in the training objective? Is the composite score used during training or only at inference? How do gradients flow through trust-related components, if at all?
- Alpha sensitivity: How sensitive are results to the alpha parameter in the composite score? Please report a sensitivity analysis and explain whether alpha is tuned per dataset, per user, or learned.
- Attention mechanism: How do you inject trust into attention? Is trust an explicit feature in the attention score, or only present as edge weights in the adjacency? Please clarify and provide the precise attention formulation when trust is present.
- Explainability evaluation: How are 'explanatory paths' constructed and ranked? Please provide quantitative metrics (e.g., explanation fidelity/coverage) or a user study, and include qualitative examples illustrating how trust changes the explanation.
- Datasets and splits: What are the exact preprocessing steps per dataset (filtering thresholds, implicit/explicit conversion), train/validation/test splits, and interaction density? Are results averaged across multiple random splits?
- Baselines: Please compare against stronger baselines including TrustSVD and recent GNN-based recommenders (e.g., NGCF, LightGCN, GraphRec, SocialGCN). If not feasible, justify why and provide at least one competitive trust-aware baseline.

- Variance and significance: Please report standard deviations or confidence intervals over multiple runs with fixed seeds, and correct the p-value reporting ('p < 0.05'?).
- Implementation details: Provide random seeds, epochs, optimizer, learning rate schedule, embedding dimension, weight decay, early stopping criteria, and hardware specs. Will you release code to support reproducibility?
- Global pooling rationale: Why is a global pooling to a graph-level representation appropriate for user–item recommendation? If you use user–item pair scoring, please specify the final scoring function (e.g., dot product, MLP over concatenated embeddings) and how it relates to the composite score.

Limitations

- Ambiguity of trust: Without a formal trust derivation, the approach risks conflating 'trust' with activity or popularity, potentially amplifying exposure bias.
- Cold-start and sparsity: Heavy reliance on historical interaction metrics may penalize new users/items or sparse users, reducing fairness and coverage.
- Temporal drift: Static trust scores may become stale; trust can be time-varying and context-dependent.
- Gaming and robustness: If users learn that activity inflates trust, the system may be gamed (e.g., review spamming). Robustness defenses are not discussed.
- Privacy: Deriving trust from detailed histories could raise privacy concerns if sensitive behavioral signals are used without careful safeguards.
- Explanation risks: Path-based explanations could provide post-hoc rationales misaligned with the actual decision process unless fidelity is measured; misleading explanations can erode trust.

Image Evaluation

Cross-Modal Consistency: 24/50

Textual Logical Soundness: 16/30

Visual Aesthetics & Clarity: 9/20

Overall Score: 49/100

Detailed Evaluation (≤500 words):

1. Cross-Modal Consistency

- Visual ground truth: No figures. Two tables: Table 1 (model performance), Table 2 (ablation). No per-dataset results or explanation visuals.
- Major 1: Table numbering mismatch; text cites “Tables 4.3 and 4.3” but only Table 1 and Table 2 exist. Evidence: “Tables 4.3 and 4.3 summarize the recommendation accuracy...” (Sec 4.3).
- Major 2: Claims four datasets, but tables report a single set of metrics without per-dataset breakdown, blocking verification. Evidence: “experiments on four diverse datasets: Yelp, Gowalla, ML-10M... Amazon” (Sec 4.1).

- Major 3: Transparency/explanation claims lack any figure/table/example paths to align with text. Evidence: “Every recommendation accompanies a path-based explanation...” (Sec 3.3).

- Minor 1: Statistical notation error reduces clarity. Evidence: “significant (p; 0.05)” (Sec 4.3).

2. Text Logic

- Major 1: Placeholder citations around core equations undermine attribution and reproducibility. Evidence: “ensuring robust... as seen in (?)” and “advancing the model’s selectivity... (?)” (Sec 3.2).

- Major 2: Statistical significance claim is unsupported (no test statistic, variance, or per-dataset p-values). Evidence: “paired t-tests... significant (p; 0.05)” (Sec 4.3).

- Minor 1: Trust derivation insufficiently specified (no concrete mapping/formula). Evidence: “Trust scores calculated from historical interaction metrics...” (Sec 3.1).

- Minor 2: Composite score is under-specified (what is Trust Score, calibration, bounds?). Evidence: “Composite Score = $\alpha \times \text{Trust Score} + (1 - \alpha) \times \text{GNN Score}$ ” (Sec 3.2).

3. Figure Quality

- Major 1: Tables omit critical identifiers (dataset names/splits) and uncertainty (std/CI), impeding interpretation of claimed improvements. Evidence: Table 1 and Table 2 contain single-point metrics only (Sec 4.3).

- Minor 1: Table captions are generic and lack methodological context (e.g., evaluation protocol, filtering rules). Evidence: “Table 1: Recommendation Performance of GTERM” (Sec 4.3).

Key strengths:

- Clear motivation to integrate trust with GNNs; simple hybrid GCN+GAT with α -weighted fusion.
- Ablation includes “w/o Trust” and “w/o GAT,” showing incremental contributions.
- Modular implementation details (PyG, DataLoader) indicate practical focus.

Key weaknesses:

- Cross-modal gaps: misnumbered tables and no per-dataset reporting despite multi-dataset claims.
- Transparency is asserted but not evidenced (no example paths, user studies, or explainability metrics).
- Placeholder citations and incomplete statistical reporting reduce credibility.
- Trust score construction lacks a precise, reproducible definition.

Recommendations:

- Report per-dataset results with variance and correct table numbering.
- Provide concrete trust computation formulas and remove “(?)”.
- Add explanation exemplars/figures and an explainability evaluation protocol.
- Detail statistical tests (test stats, p-values per dataset) and evaluation splits.

Scores

Originality: **2**

Quality: **2**

Clarity: **2**

Significance: **2**

Soundness: 2

Presentation: 2

Contribution: 2

Rating: 3

AI Review from SafeReviewer

AI Review from SafeReviewer will be automatically processed

Summary

This paper introduces GTERM, a novel recommendation framework that aims to enhance the transparency and trustworthiness of Graph Neural Network (GNN)-based recommendation systems by incorporating trust metrics. The core idea is to augment the user-item interaction graph with trust information, derived from historical interactions, and then use a combination of Graph Convolutional Networks (GCN) and Graph Attention Networks (GAT) to learn node embeddings. The model calculates a composite score by combining interaction predictions from the GNN with the trust scores, intending to provide more accurate and interpretable recommendations. The authors claim that GTERM improves recommendation accuracy and provides transparent, trust-enhanced recommendations, thereby increasing user confidence and satisfaction. Empirically, the paper evaluates GTERM on four benchmark datasets, comparing its performance against basic Collaborative Filtering (CF) and a standard GCN model. The results, as presented, suggest that GTERM outperforms these baselines in terms of Recall, Precision, and NDCG. The paper also includes an ablation study to demonstrate the impact of the trust mechanism and the GAT component. While the paper presents an interesting approach to integrating trust into GNN-based recommendations, my analysis reveals several critical limitations in its methodology, experimental design, and presentation. The lack of novelty in the core idea, the superficial integration of trust, the inadequate experimental evaluation, and the poor writing quality significantly undermine the paper's overall contribution. Despite the potential of the research direction, the current implementation falls short of providing a robust and convincing argument for the proposed model's effectiveness and novelty.

Strengths

The paper's exploration of integrating trust mechanisms into GNN-based recommendation systems is a valuable and timely research direction. The idea of using trust to enhance the transparency and reliability of recommendations is well-motivated, and the potential benefits of such an approach are significant. The authors correctly identify the limitations of traditional GNN-based recommenders, which often overlook the importance of trust in user-item interactions. The use of a composite score, combining GNN predictions with trust scores, is a reasonable attempt to incorporate trust into the recommendation process. Furthermore, the paper's inclusion of an ablation study, even though limited, demonstrates an effort to understand the contribution of different components of the model. The experimental results, while not entirely convincing due to the choice of baselines, do suggest that the proposed model can achieve some performance gains compared to basic CF and a standard GCN. The paper also attempts to address the explainability of recommendations by providing path-based

explanations, which is a crucial aspect of transparent recommendation systems. Despite the significant weaknesses I have identified, the paper's core idea and its attempt to integrate trust into GNNs represent a potentially fruitful avenue for future research.

Weaknesses

My analysis reveals several significant weaknesses that undermine the paper's contribution. First, the novelty of the proposed method is limited. While the paper introduces GTERM as a novel approach, the core idea of incorporating trust into recommendation systems is not new, as the authors themselves acknowledge in the related work section. Furthermore, the use of a composite score to combine GNN predictions with trust scores mirrors existing approaches in the literature, as pointed out by one of the reviewers. The paper fails to adequately differentiate its approach from existing trust-enhanced recommendation models, particularly those using GNNs. The lack of a detailed comparison with these models in the related work section is a critical oversight. Second, the integration of trust into the GNN framework is superficial. The paper calculates trust scores based on simple heuristics like review frequencies and ratings, without exploring more sophisticated methods for trust derivation. The trust scores are then used as edge weights in the graph and combined with the GNN output using a simple weighted sum. This approach does not deeply integrate trust into the GNN's learning process. The paper does not explore how the GNN architecture itself can be modified to better incorporate trust information, such as using trust-aware attention mechanisms or trust-based message passing. This lack of deep integration limits the potential of the model to truly leverage the power of GNNs for trust-enhanced recommendations. Third, the experimental evaluation is inadequate. The paper compares GTERM against only basic CF and a standard GCN model. It fails to include comparisons with more recent and sophisticated GNN-based recommendation models, including those that also incorporate trust or similar mechanisms. This lack of strong baselines makes it difficult to assess the true effectiveness and novelty of GTERM. The experimental results, while showing some improvement over the baselines, are not convincing due to the weak comparisons. The paper also lacks a detailed analysis of the impact of the trust mechanism on the model's performance. The ablation study, while helpful, does not provide a comprehensive understanding of how trust affects different aspects of the model's behavior. Fourth, the paper's writing quality is poor. The method description is vague and lacks necessary details. The paper uses undefined symbols in equations, making them difficult to understand. The explanation of the path-based explanations is also unclear. The overall presentation of the paper is not well-organized, and the language is often imprecise. The introduction does not clearly articulate the research gap, and the related work section does not provide a comprehensive overview of the existing literature. Finally, the paper lacks a thorough discussion of the ethical implications of using trust in recommendation systems. The paper does not address the potential for bias in the trust metrics and how this could affect the recommendations. The paper also does not discuss the potential for misuse of trust information and how to mitigate these risks. These ethical considerations are crucial for the responsible development of trust-enhanced recommendation systems. All of these weaknesses are supported by direct evidence from the paper and significantly impact the paper's overall credibility and contribution.

Suggestions

To address the identified weaknesses, I recommend several concrete improvements. First, the authors should clearly articulate the novelty of their approach by providing a detailed comparison with existing trust-enhanced GNN-based recommendation models. This comparison should highlight the specific limitations of existing models that GTERM aims to address and clearly define the unique contributions of their work. The authors should also explore more sophisticated methods for trust derivation, going beyond simple heuristics like review frequencies and ratings. This could involve incorporating social network information, user behavior patterns, or temporal dynamics into the trust calculation. Second, the authors should deeply integrate trust into the GNN architecture. This could involve developing trust-aware attention mechanisms, trust-based message passing, or other methods that allow the GNN to learn from trust information more effectively. The simple weighted sum approach should be replaced with a more nuanced integration strategy. Third, the experimental evaluation should be significantly improved. The authors should include comparisons with a wider range of baselines, including more recent and sophisticated

GNN-based recommendation models, as well as state-of-the-art trust-aware recommendation systems. The experimental results should be presented with more detail, including statistical significance tests and visualizations. The authors should also conduct a more thorough analysis of the impact of the trust mechanism on the model's performance, exploring how trust affects different aspects of the model's behavior. Fourth, the paper's writing quality needs substantial improvement. The authors should provide a clear and detailed description of their methodology, including all necessary mathematical notations and definitions. The paper should be well-organized, with a clear introduction, related work section, methodology section, experimental section, and conclusion. The language should be precise and concise. The authors should also provide a more detailed explanation of the path-based explanations. Finally, the authors should include a thorough discussion of the ethical implications of using trust in recommendation systems. This discussion should address the potential for bias in the trust metrics, the potential for misuse of trust information, and how to mitigate these risks. The authors should also consider the potential impact of their model on user privacy and data security. These improvements would significantly strengthen the paper and make a more convincing argument for the proposed model's effectiveness and novelty.

 Questions

Based on my analysis, I have several questions that I believe are crucial for understanding the paper's contributions and limitations. First, how does the proposed model specifically address the limitations of existing trust-enhanced GNN-based recommendation models? The paper does not provide a detailed comparison with these models, making it difficult to assess the unique contributions of GTERM. Second, what is the rationale behind the specific method used for trust score derivation? The paper mentions using review frequencies and ratings, but does not provide a detailed justification for this choice. Are there other, more sophisticated methods for trust derivation that could be explored? Third, why was a simple weighted sum used to combine the GNN predictions with the trust scores? What is the justification for this approach, and how does it compare to other methods for combining these two sources of information? Fourth, how does the model ensure that the trust scores are not biased or manipulated? The paper does not address the potential for bias in the trust metrics, which could lead to unfair or discriminatory recommendations. Fifth, how are the path-based explanations generated, and what is the evidence that these explanations are meaningful to users? The paper does not provide a clear description of the explanation generation process, nor does it include any user studies to validate the quality of the explanations. Finally, what are the computational costs of the proposed model, and how does it scale to large datasets? The paper does not provide any analysis of the model's computational complexity or its performance on large-scale datasets. These questions target the core methodological choices and assumptions of the paper, and addressing them would significantly improve the paper's overall credibility and contribution.

 Scores

Soundness: 1.75

Presentation: 1.5

Contribution: 1.25

Confidence: 4.25

Rating: 2.0

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Insights

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